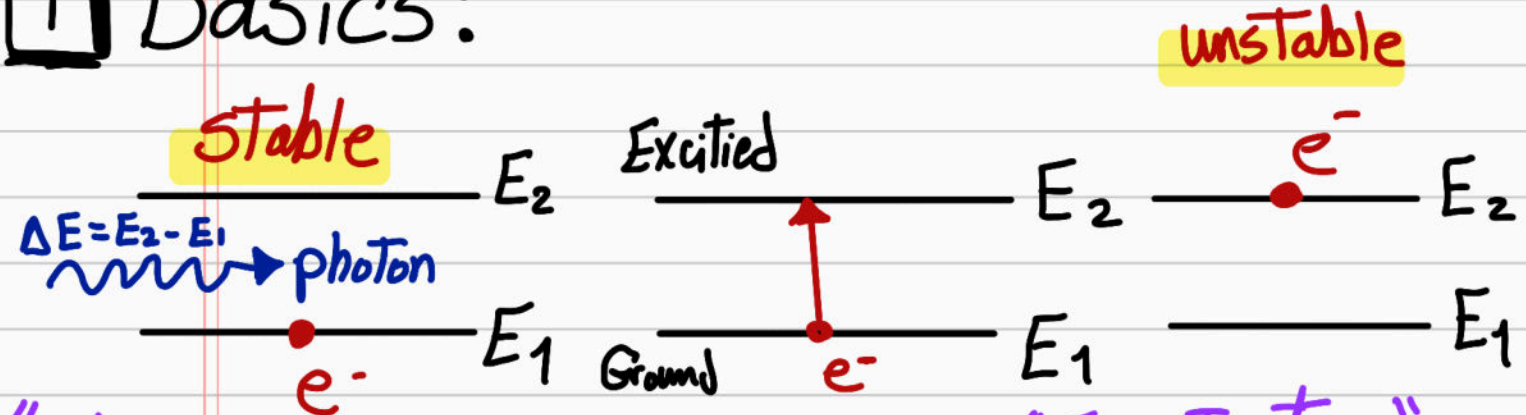
L.A.S.E.R.

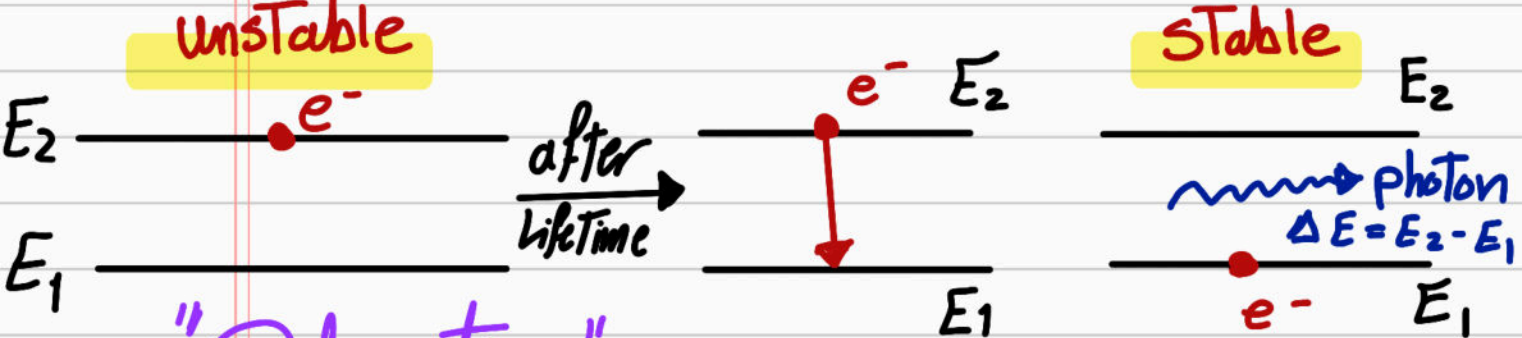
Light **A**mplification by **S**timulated **E**mission **R**adiation

تقوية تحفيز

1 Basics:



"Photon Absorption" ⟶ "Excitation"



"Relaxation" ⟶ "photon Emission"

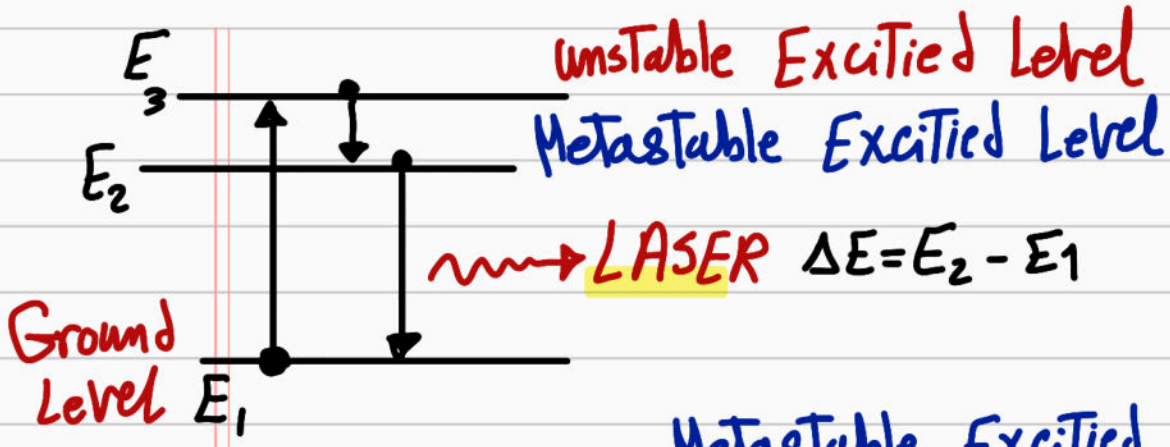
LifeTime

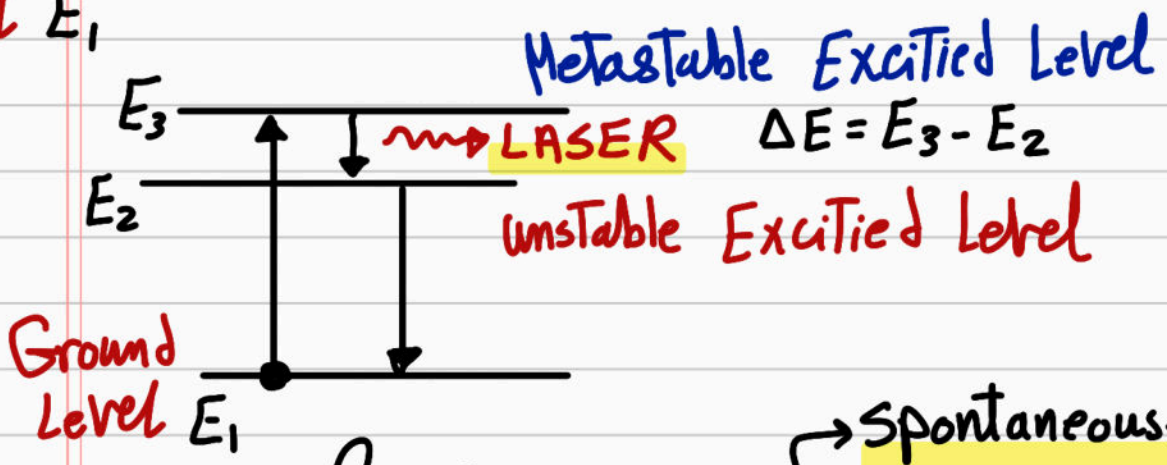
Meta-stable Excited Level — Relatively longer LifeTime

10^{-3} Second = 0.001 second

unstable Excited Level — Shorter LifeTime

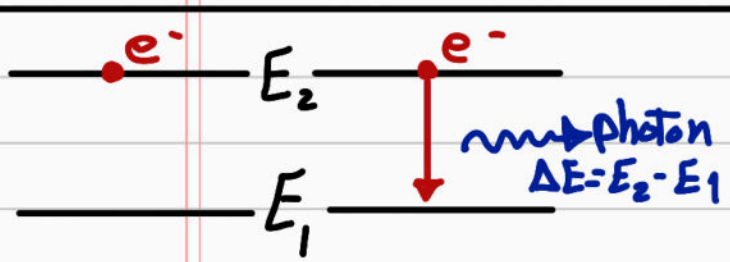
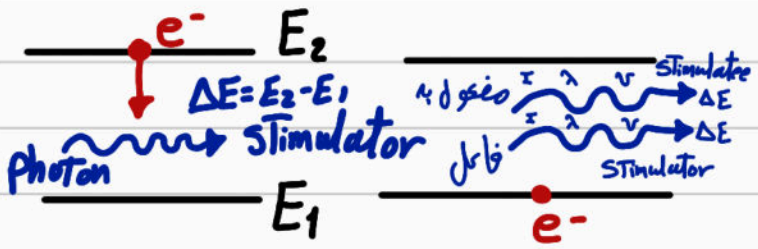
10^{-8} Second = 0.00000001 sec.



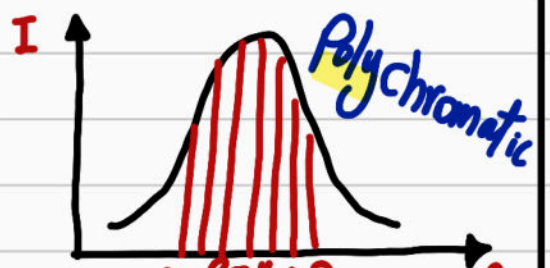
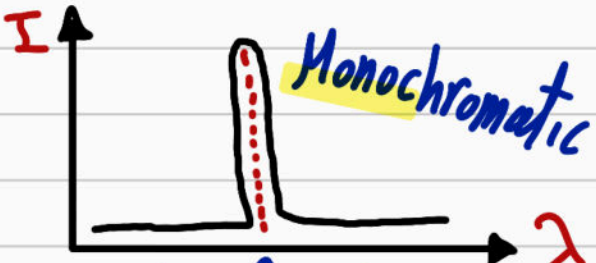
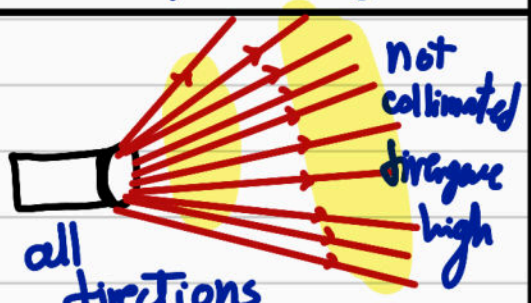
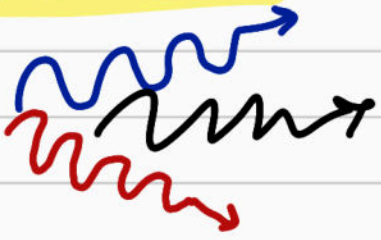
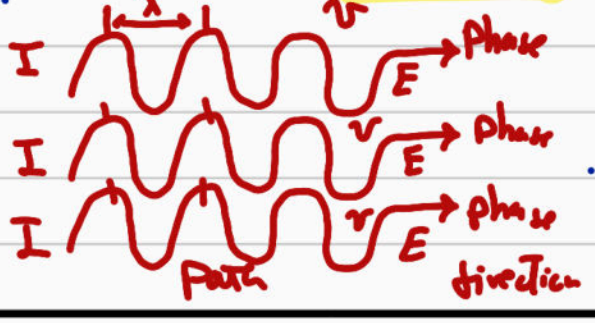
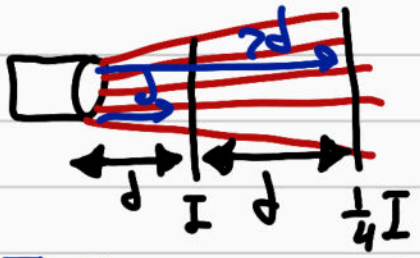
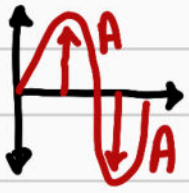
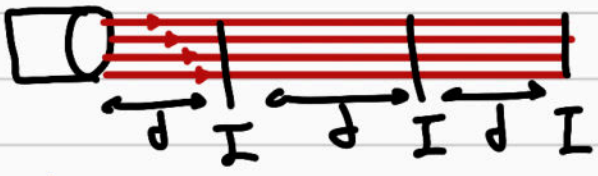
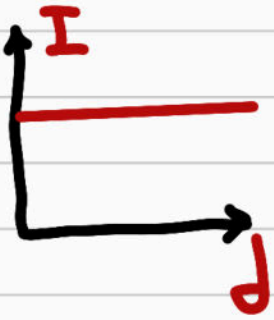


2 Types of Emission

- Spontaneouse — تلقائي
- Stimulated — تحفيز

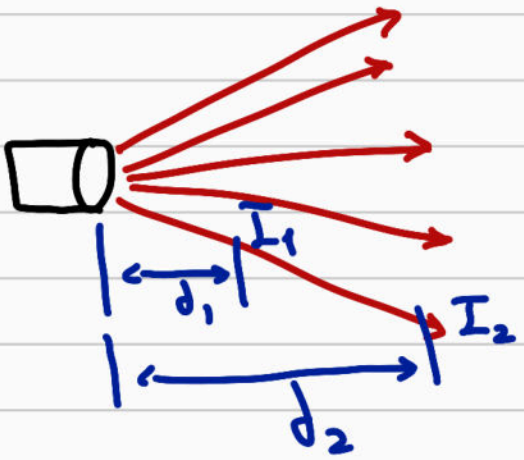
Spontaneouse Emission	Stimulated Emission
1. After Life Time 2. Without using external stimulator	1. Before LifeTime 2. Using external stimulator
$t \geq$ LifeTime	$t <$ LifeTime
	
Ordinary light 1. Tungestn blub 2. Neon lamp 3. Xenon lamp	LASER

3 Light Vs LASER

P.O.C	Light Spontaneous Emission	LASER STimulated Emission
Monochromatic 1 λ Spectral purity	 Band of Wavelengths	 Single specific Wavelength
Collimation		Parallel lines collimated  in one direction
Coherency	Photons are incoherent 	Photons are coherent 
Intensity	 Inverse square law $I \propto \frac{1}{d^2} \rightarrow \frac{I_1}{I_2} = \left(\frac{d_2}{d_1}\right)^2$ $I \propto \text{Amplitude}^2$ $\frac{I_1}{I_2} = \left(\frac{A_1}{A_2}\right)^2$ 	 doesn't obey Inverse Square 

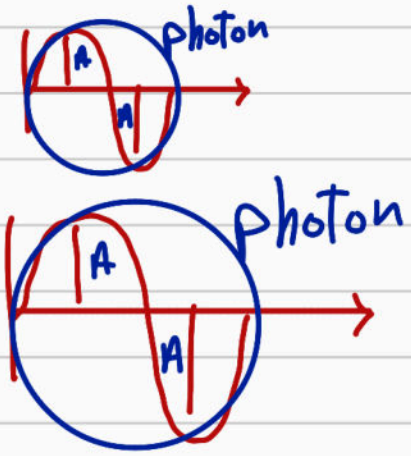
Light Rule

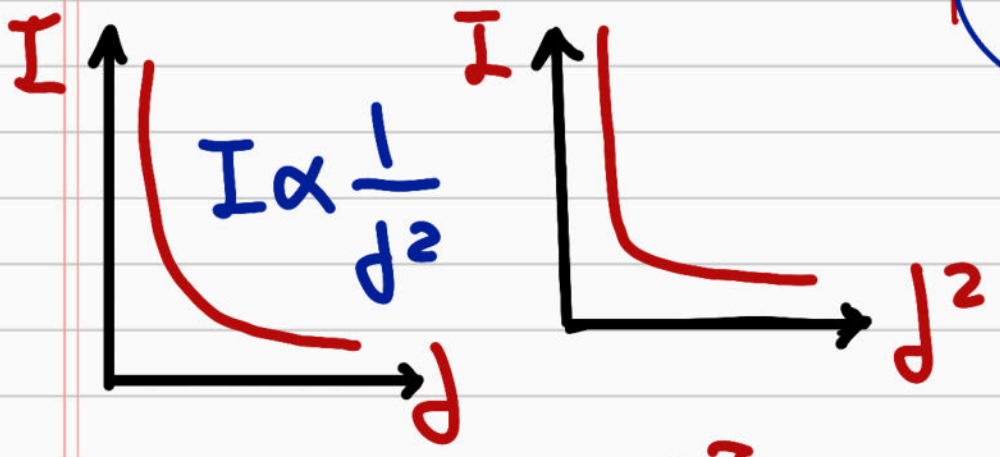
$$\frac{I_1}{I_2} = \left(\frac{d_2}{d_1}\right)^2 = \left(\frac{A_1}{A_2}\right)^2$$

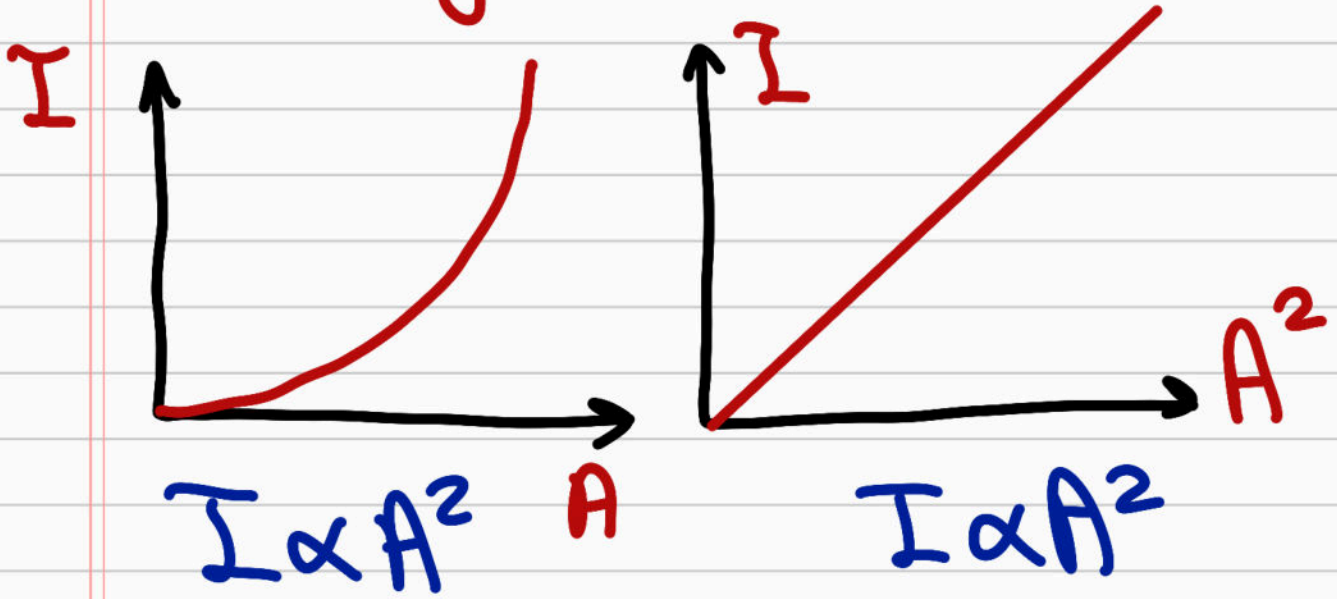


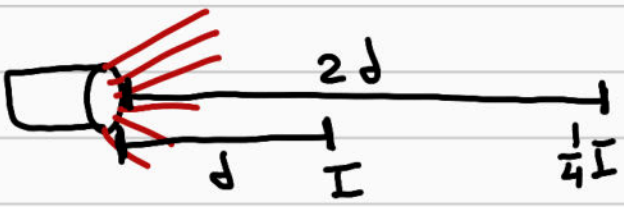
A → Amplitude

$$\sqrt{\frac{I_1}{I_2}} = \frac{d_2}{d_1} = \frac{A_1}{A_2}$$









$$\frac{I_1}{I_2} = \left(\frac{d_2}{d_1}\right)^2 \rightarrow \frac{I}{I_2} = \left(\frac{2d}{d}\right)^2 = \frac{4}{1}$$

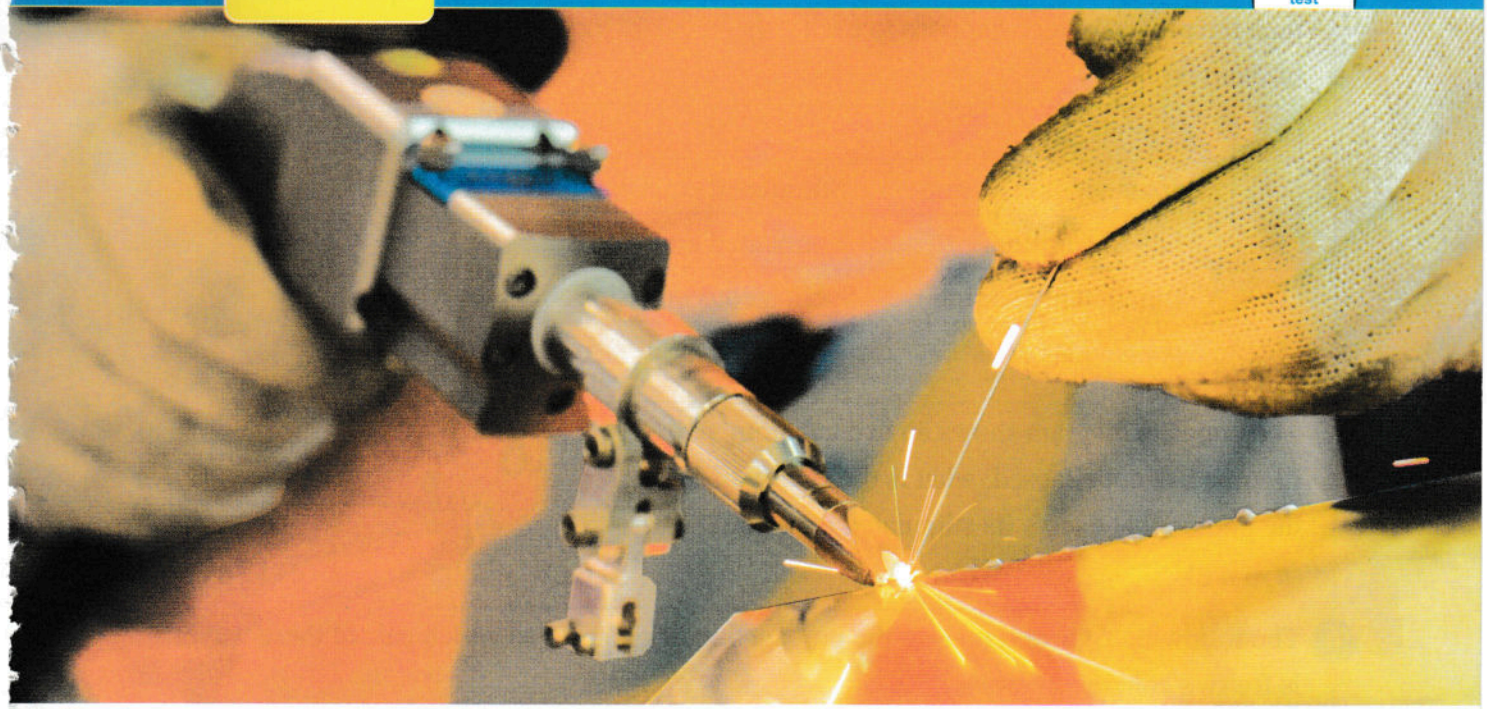
$$\frac{I}{I_2} = \frac{4}{1} \rightarrow I_2 = \frac{1}{4}I$$

Chapter 7

Questions on Chapter Seven

Laser

Interactive test



The questions signed by ✱ are answered in detail. ● Understand ○ Apply Higher Order Thinking Skills

First Multiple choice questions

1 In the shown light source, is the dominant emission spontaneous or stimulated?

(a) Spontaneous emission. ✓
(b) Stimulated emission.
(c) Both of them occurs in the same proportion.
(d) The information is not enough to determine the answer.



Ordinary light

2 The ratio between the lifetime of an atom in the unstable excitation state and the lifetime of the atom in the metastable state is ..... .

10^{-8} s, 10^{-3}

$\frac{10^{-8}}{10^{-3}} = 10^{-5} = 0.00001 < 1$

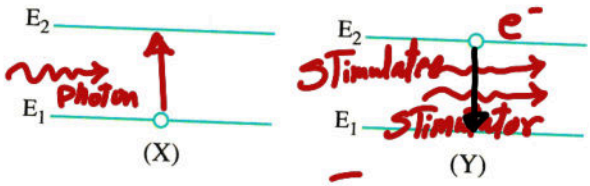
(a) greater than 1 (b) equal to 1
(c) less than 1 ✓ (d) the information is not sufficient for determining the answer

3 In the neon bulb, ...



(a) the dominant emission is the photoelectric emission
(b) the dominant emission is the spontaneous emission ✓
(c) the dominant emission is the stimulated emission
(d) the stimulated and the spontaneous emissions have the same proportions

4 In the opposite figure, when a photon of energy $(E_2 - E_1)$ passes through two atoms (X) and (Y) of an active medium, then the process which happens to each of them is ..... .



	X	Y
(a)	Spontaneous emission	Stimulated emission
(b)	Absorption	Stimulated emission
(c)	Stimulated emission	Spontaneous emission
(d)	Spontaneous emission	Absorption

$E_2 - E_1$

$\Delta E_2 = -3.4 - (-13.6)$

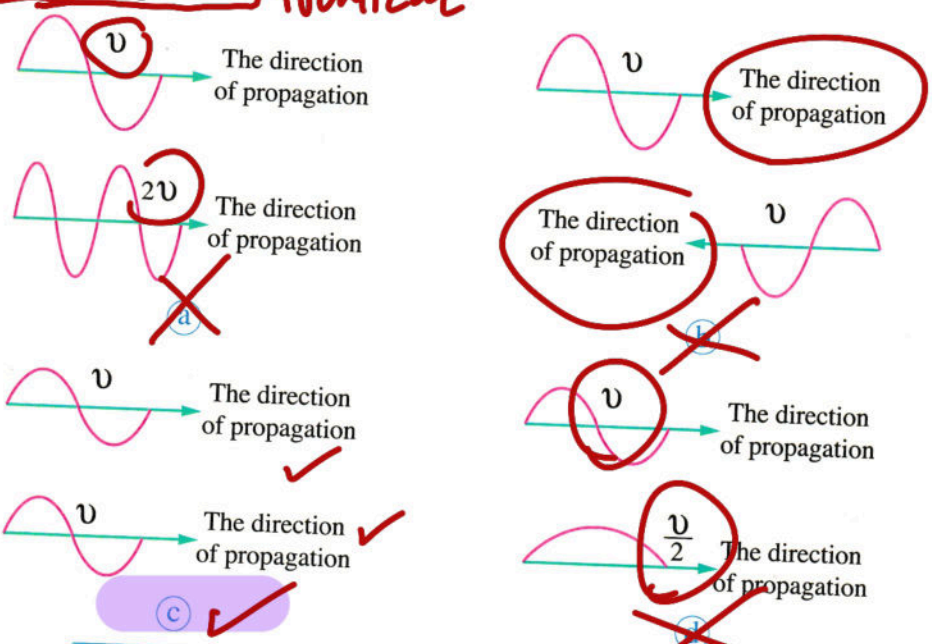
5 The opposite diagram shows an excited atom whose lifetime has not ellapsed, what is the energy of the photon that causes its stimulated emission?

$\Delta E = E_2 - E_1$



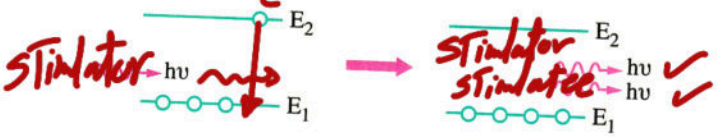
(a) 3.4 eV (b) 10.2 eV (c) 13.6 eV (d) 17 eV

6 The following figures represent the propagation of photons, so which pair of these waves is for two coherent photons? identical

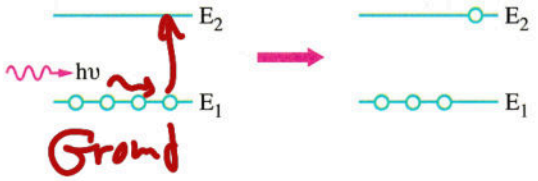


7 The opposite diagram represents the process of ..... .

(a) spontaneous emission
(b) spontaneous absorption
(c) stimulated emission
(d) stimulated absorption

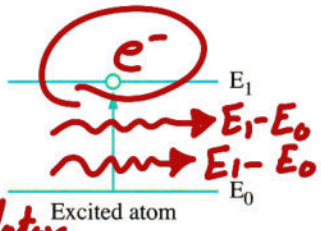


8 The opposite diagram represents the case of ..... .



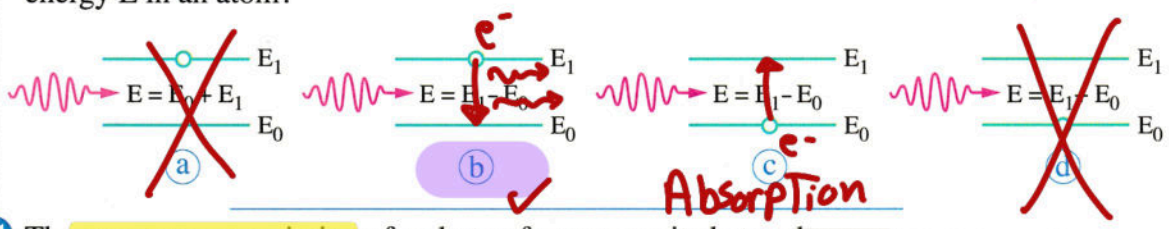
(a) a spontaneous emission
(b) a stimulated emission
(c) an absorption
(d) a population inversion

9 The opposite diagram shows an excited atom in energy level E_1, so which of the following sentences clarifies the required condition for achieving stimulated emission from this atom?



(a) Elapsing of the lifetime for the atom in level E_1.
(b) The collision of a free electron of energy $(E_1 - E_0)$ with the atom.
(c) The incidence of a photon of energy $(E_1 - E_0)$ on the atom.
(d) The collision of another excited atom in level E_1 with this atom.

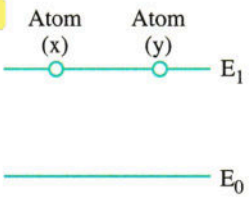
10 Which of the following cases represents a case of stimulated emission by a photon of energy E in an atom?



11 The spontaneous emission of a photon from an excited atom happens ..... .

(a) when a photon falls on the excited atom
(b) under the effect of a low frequency photon
(c) without any external effect
(d) under the effect of a high frequency photon

12 In the opposite diagram, two atoms (x) and (y) of the same element were excited to the same energy level (E). Upon returning to their ground state (E_0), atom (x) emits a photon through spontaneous emission, and atom (y) returns due to the influence of an external photon and emits a photon through stimulated emission, so in which of the following do atoms (x) and (y) differ?

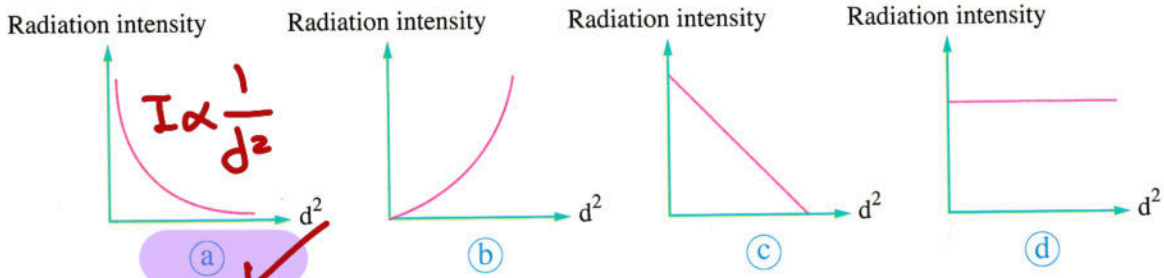


(a) The energy of the emitted photon.
(b) The time the atom spends at level (E_1).
(c) The speed of the emitted photon.
(d) All of the above are identical for both atoms.

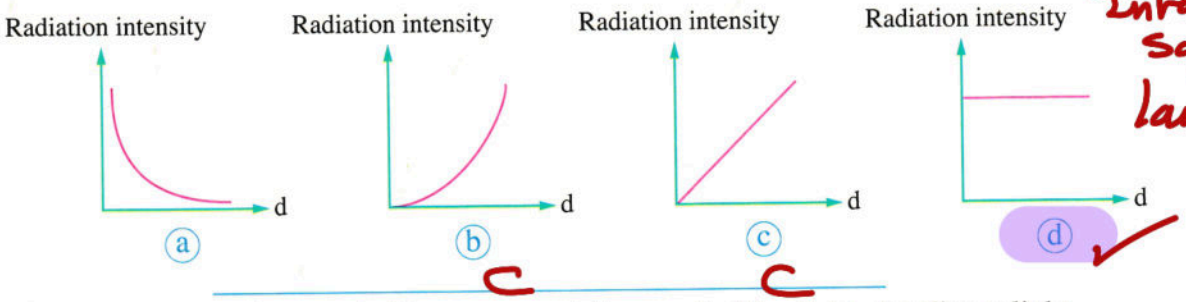
13 Two atoms (1), (2) of the same element are excited to an energy level of lifetime 10^{-3} s, if a photon is emitted from each atom after being in that level for an interval of 10^{-5} s, 10^{-3} s respectively, so the photon emitted from :

	Atom (1)	Atom (2)
(a)	by an external photon	by an external photon
(b)	by an external photon ✓	without external stimulus ✓
(c)	without external stimulus	by an external photon
(d)	without external stimulus	without external stimulus

14 The graph which represents the relation between the radiation intensity of the light of an electric lamp and the square of the distance (d^2) travelled by this light away from the lamp is ..... .



15 The graph which represents the relation between the radiation intensity of the light of a laser source and the distance (d) travelled by this light away from the laser is .....



16 The ratio between the speed of laser rays and the speed of the light of ordinary light sources in space is ..... .

(a) greater than one
(b) less than one
(c) equal to one ✓
(d) the answer can't be determined

17 The common feature between laser and X-rays photons is that they ..... .

(a) are coherent
(b) are monochromatic
(c) have the same speed ✓
(d) have the same energy

18 If a surface is exposed to different light sources having the same light power which are placed at the same distance from the surface, the intensity of light on the surface will be higher when using ..... .

(a) the light of the tungsten lamp (b) the light of the fluorescent lamp

(c) the light of the neon lamp (d) the light of laser

19 Laser rays don't obey the inverse square law as ordinary light rays because their photons .....

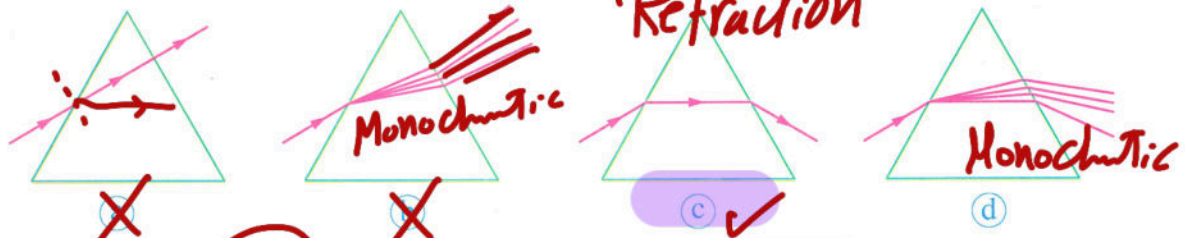
(a) are coherent (b) are scattered on the air particles

(c) have one wavelength (d) have a large divergence angle

20 The laser rays in the same beam are parallel, so, their photons have the same ..... .

(a) direction (b) frequency (c) intensity (d) wavelength

21 Which of the following figures represents the passage of helium-neon laser through an equilateral triangular prism at minimum deviation?

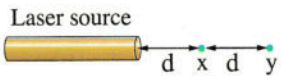


(a) (b) (c) (d)

22 If the intensity of a laser beam at a distance of 10 cm from its source is I, its intensity at a distance of 20 cm is .....

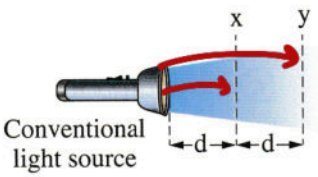
(a) 2 I (b) I (c) $\frac{I}{2}$ (d) $\frac{I}{4}$

23 If the laser source shown in the opposite figure is operated, the ratio of the radiation intensity at x to that at y $\left(\frac{I_x}{I_y}\right)$ is ..... .



(a) $\frac{1}{1}$ (b) $\frac{4}{1}$ (c) $\frac{1}{4}$ (d) $\frac{2}{1}$

24 The opposite figure shows the propagation of the rays of a conventional light source, so the ratio between the amplitude of light wave at point y and the amplitude of light wave at point x $\left(\frac{A_y}{A_x}\right)$ equals ..... .



(a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{1}{8}$ (d) $\frac{1}{16}$

4 LASER Components: ASR

1 Active mediume:

* Solid → Crystal → Ruby الياقوت
  → semi-conductor → Silicon
* Liquid → organic dyes → dissolve in Water
* Gas → Atoms H + He
  → ionized Ar or He-Ne
  → molcules Co_2

2 Source of energy:

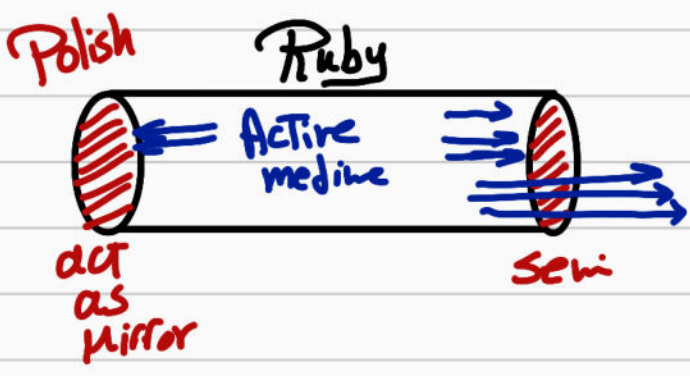
* Electric Energy → Dc, Ac } Gas
* Optical Energy (pumping) (ضوء)
  → flash light → Ruby
  → LASER → organic dyes
* Thermal Energy → Pressure Transfer from one area To another
* chemical Energy → Exothermic → heat + Energy
  "H + deutrim Floride + F + Co_2"

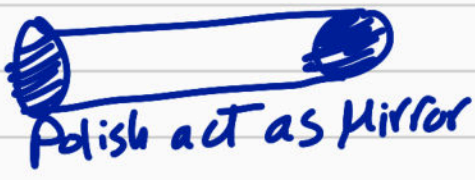
3 Resonant caviTiy: Amplification

* External



* Internal



P.O.C.	He - Ne	Ruby
Active medium	Gas He : Ne	Solid Crystal Ruby
Source of energy	Electric Energy	Optical Energy (flash lamp)
Resonant Cavity	External 	Internal 

5 LASER Action Theory:

Source —Excite→ Active med. —Reaches→ Unstable energy Level

$e^-_{Meta} > e^-_{Ground}$

After LifeTime 10^{-8}s ↓

Metastable energy Level 10^{-3}s —electrons accumulates تتراكم→ Population inversion —After LifeTime 10^{-3}s→ Spont. Emission

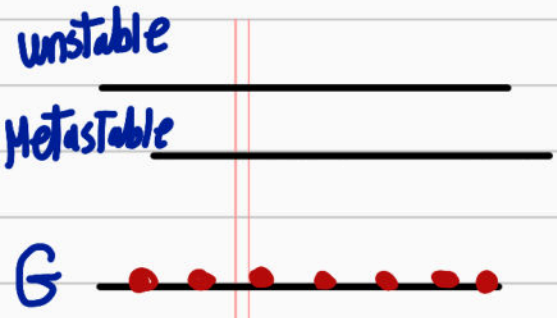
Before Lifetime 10^{-3}s ↓

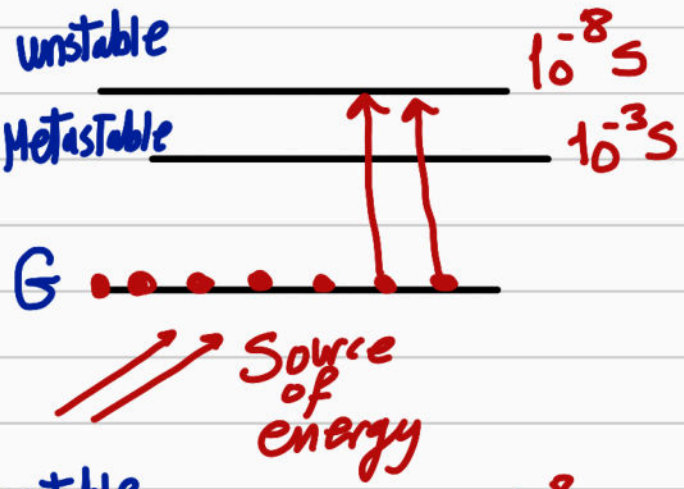
Stimulated Emission —Try To escape→ Mirrors —Reflect back→ Resonant cavity

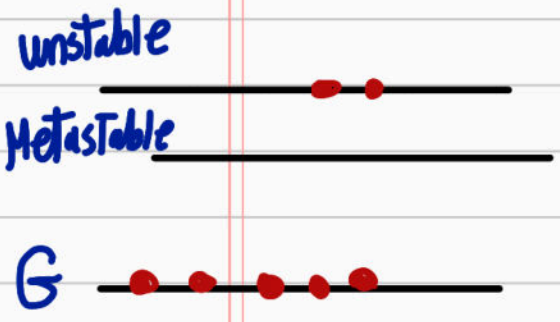
Multiple Reflections ↓

Amplification process (no. of photons ↑↑) —Certain intensity (Mirrors Reflection coff.)→ LASER

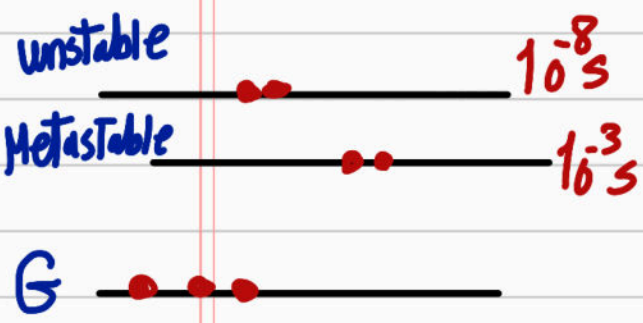
Semi Transparat

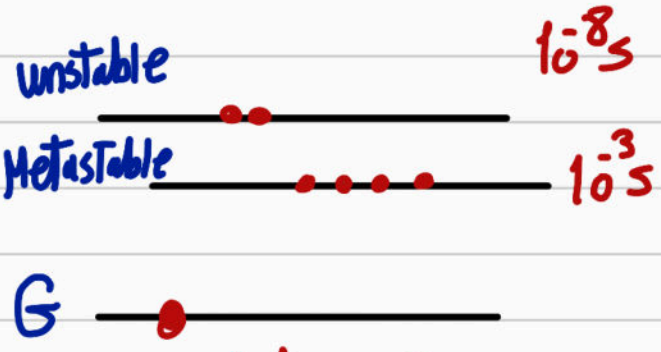




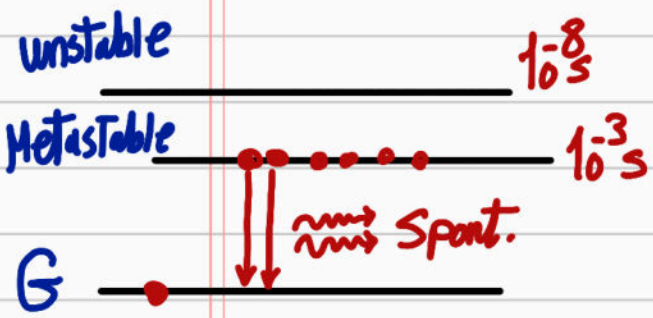


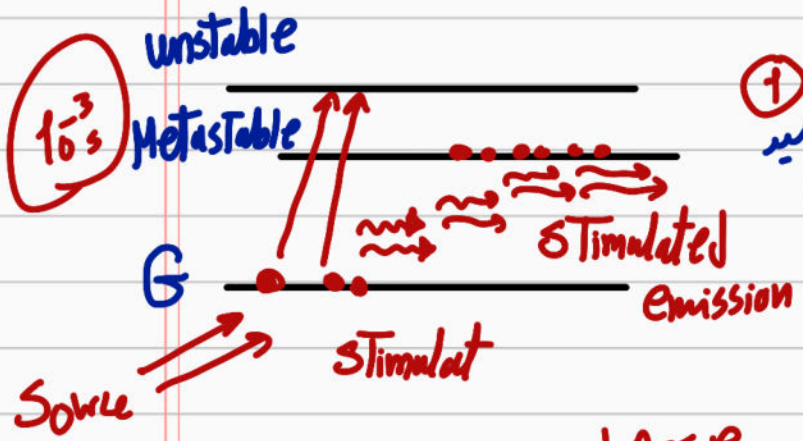


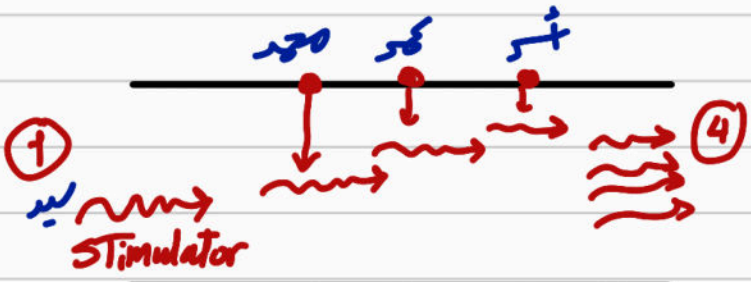


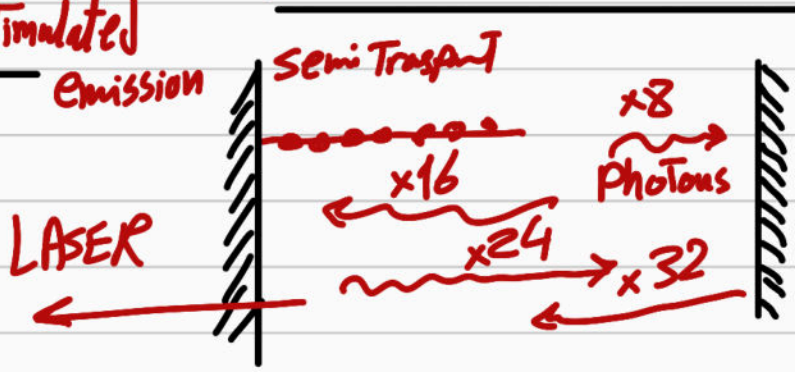


→ Population inversion ←
no. of electrons in Metastable state is greater than no. of electrons in ground state







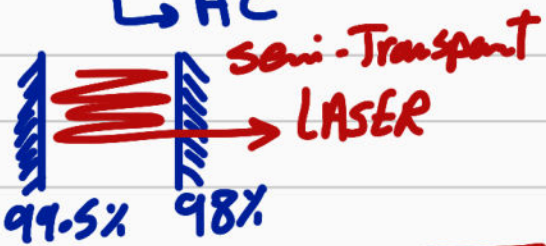


6 He-Ne LASER:

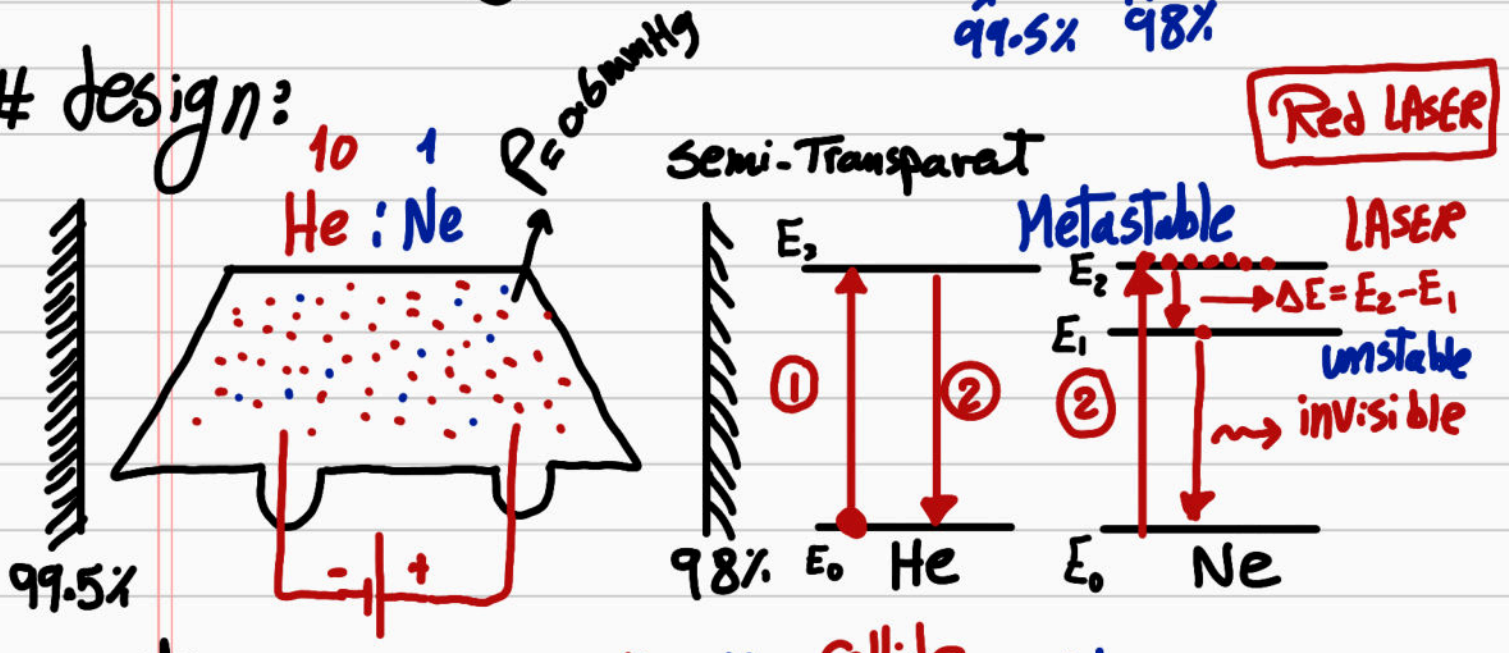
#Active medium: He : Ne (10 : 1) Pressure (Very Low pressure) 0.6 mmHg

#Source of energy: Electric Source → Dc / Ac

#Resonant cavity: External

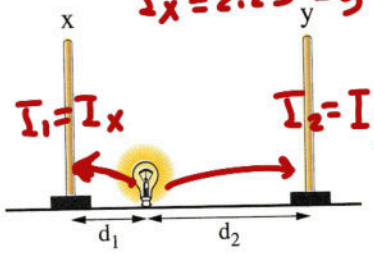


#design:



#operation: Source → (Excite) He → (collide / Excite) Ne → Reaches Metastable energy Level (10^{-3} s) → (electrons accumulates تتجمع) Population inversion ($E_{Meta} > E_{Ground}$) → (After LifeTime 10^{-3} s) Spont. Emission → (Before Lifetime 10^{-3} s) Stimulated Emission → (Try To escape) Mirrors → (Reflect back) Resonant cavity → Multiple Reflections → Amplification process (no. of photons ↑↑) → (Certain intensity; Mirrors Reflection coff.) LASER (Semi Transparat)

25 The opposite figure shows two identical flat screens (x, y) placed at different distances (d_1, d_2) beside a light source, so if the illumination intensity on screen (x) is 2.25 times as large as the illumination intensity on screen (y), the ratio of $\left(\frac{d_1}{d_2}\right)$ equals .....



(a) $\frac{1}{2}$ (b) $\frac{4}{9}$ (c) $\frac{2}{3}$ (d) $\frac{3}{4}$

26 The monochromaticity of laser rays is due to the fact that .....

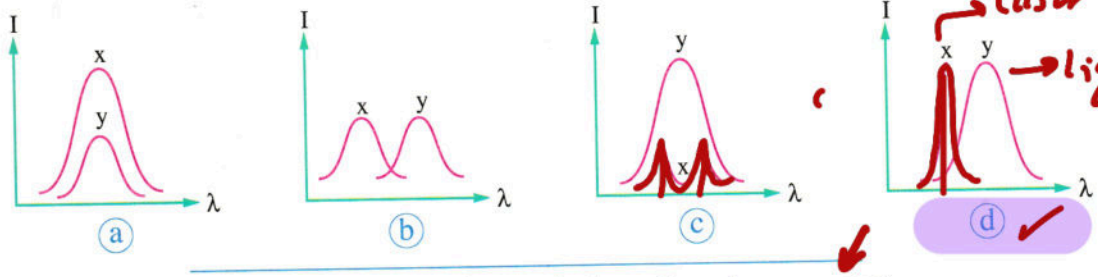
(a) all the atoms of the active medium get excited to one unstable energy level

(b) all the atoms of the active medium are in the ground energy level

(c) all the emitted photons by stimulated emission have the same energy

(d) all the emitted photons are amplified when they pass between the two reflecting mirrors

27 Two spectra (x) and (y), spectrum (x) is laser and spectrum (y) is ordinary light. Each one is analysed individually by a spectrometer, which of the following graphs represents the relation between the radiation intensity and wavelengths (λ) of the spectrum?



28 The following statements are correct for producing a laser beam except .....

(a) the spontaneous emission takes place in some atoms during the production process

(b) the intensity of a laser beam changes according to the refractive index of the semi-transparent mirror

(c) the production of laser doesn't require an external source of energy

(d) the active medium atoms in a laser contain a metastable state

29 The optical pumping process in the laser aims to achieve .....

(a) the stable state (b) the population inversion state

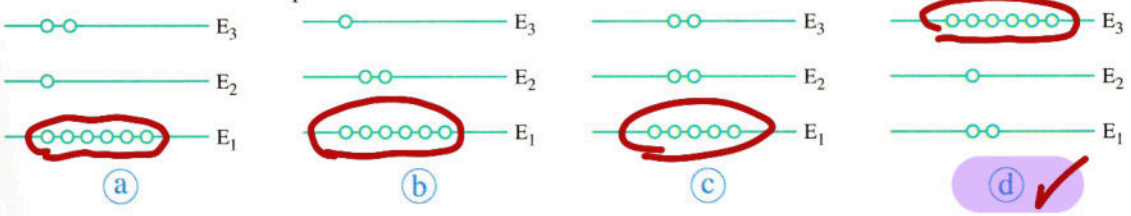
(c) the equilibrium state (d) the amplification of the laser beam

30 In laser action, the step that follows the pumping process is the achievement of .....

(a) the stable state of atoms (b) the population inversion state

(c) the equilibrium state of atoms (d) the amplification of the laser ray

31 The following diagrams show the distribution of active medium atoms among their energy levels, which of these diagrams could represent reaching population inversion? (Where energy level E_1 is the ground level).



32 The optical pumping process is used in the ..... .

(a) carbon dioxide laser (b) helium-neon laser
(c) hydrogen-fluorine laser (d) ruby laser

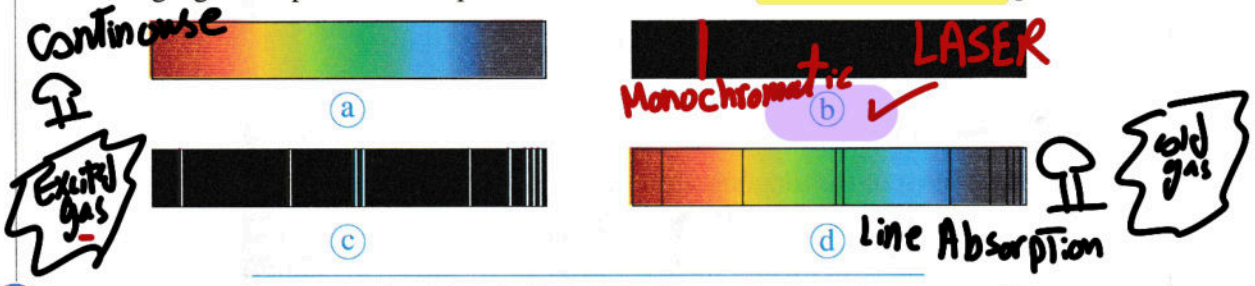
33 The process of optical pumping with a laser beam is used to produce laser whose active medium is ..... .

(a) gas atoms (b) a crystalline solid (c) a semiconductor (d) a liquid dye

34 Helium-neon laser lies in the region of .....

(a) infrared rays (b) ultraviolet rays (c) visible light (d) X-rays

35 When a spectrometer is used to analyze the emitted light from different sources, which of the following figures represents the spectrum obtained from a helium-neon laser light?



36 The laser rays photons emitted in a helium-neon laser are from the atoms of ..... .

(a) helium (b) neon
(c) both helium and neon (d) quartz

37 In helium-neon laser, to achieve the laser action, it is required to .....

(a) increase the pressure of the gases inside the tube more than the atmospheric pressure
(b) decrease the potential difference of the source
(c) increase the percentage of the atoms of the helium gas more than that of the neon gas
(d) illuminate the tube with neon light

38 The state of population inversion in helium-neon laser occurs in the atoms of ..... .

(a) helium only (b) neon only
(c) both helium and neon (d) sometimes helium and other times neon

39 In helium-neon laser, from the steps of the laser production that the excited helium atom loses its excitation energy when it collides with ..... .

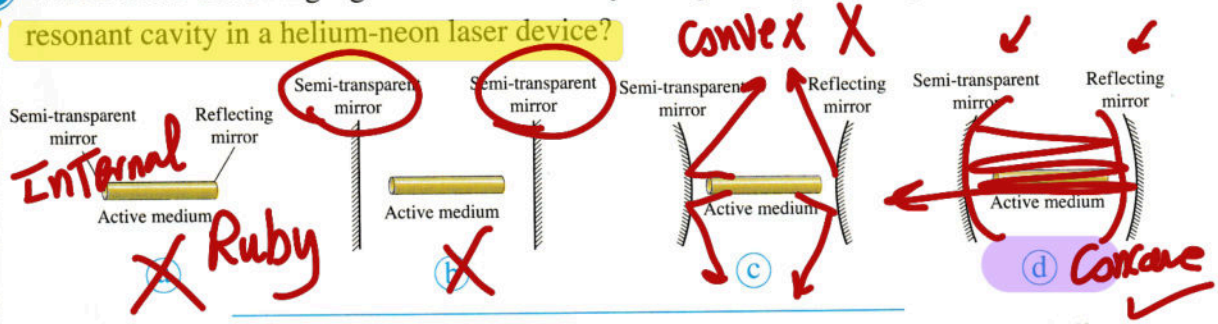
(a) another stable helium atom

(b) the walls of the electric discharge tube

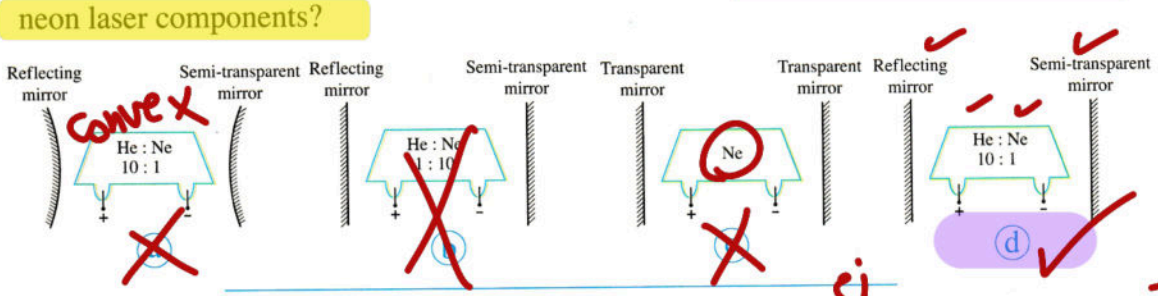
(c) an unexcited neon atom

(d) an excited helium atom

40 Which of the following figures shows correctly a diagram representing the external resonant cavity in a helium-neon laser device?



41 Which of the following diagrams represents correctly a schematic structure of a helium-neon laser components?



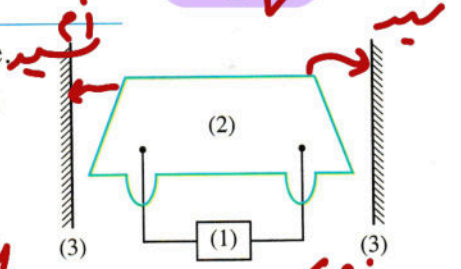
42 The opposite diagram represents a helium-neon laser device. Which component of the diagram represents the component that is responsible for the amplification process?

(a) Component (1).

(b) Component (2).

(c) Component (3).

(d) Components (1), (2).



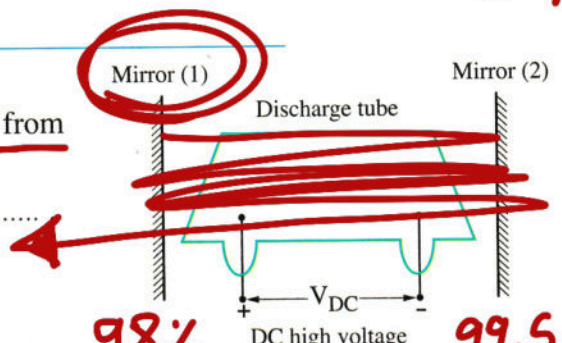
43 The opposite figure shows the structure of helium-neon laser, if the laser beam is emitted from mirror (1), then the reflection coefficient of mirror (1) compared to that of mirror (2) is .....

(a) greater

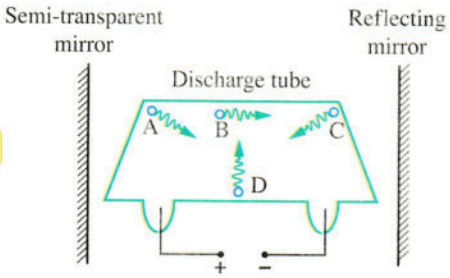
(b) less

(c) equal

(d) cannot be determined



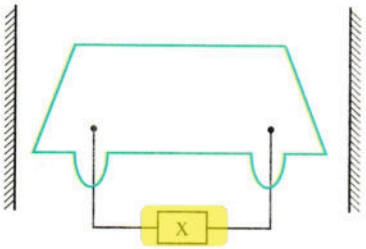
44 The opposite schematic diagram shows a helium-neon gas laser in which four photons (A, B, C and D) are emitted in different directions inside the tube, which of these photons may continue to move inside the tube for the longest time interval before it emerges?



(a) Photon A. (b) Photon B. ✓

(c) Photon C. (d) Photon D.

45 The opposite figure represents a helium-neon laser, so when component (x) stops working, ...



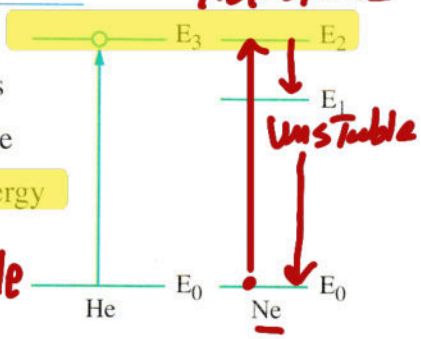
(a) the emitted radiation intensity decreases

(b) the radiation frequency decreases

(c) the emitted radiation speed decreases

(d) no laser beam is generated ✓

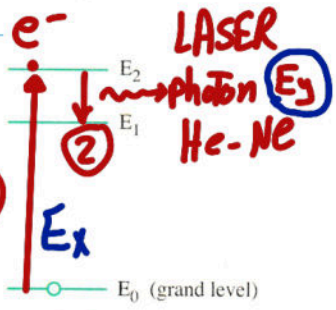
46 The opposite diagram represents the energy levels of the active medium in the (He-Ne) laser when the helium atoms in the energy level E_2 (metastable energy level) collide with the unexcited neon atoms, so the neon atoms get excited to the energy level ..... until the population inversion state is achieved.



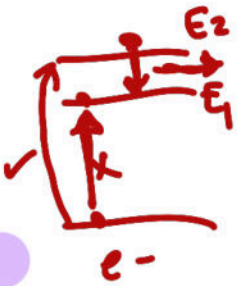
(a) E_0 only (b) E_1 only

(c) E_2 only ✓ (d) E_1 and E_2 together

47 The opposite diagram represents the energy levels of a neon atom. When the atom gained energy (E_x) it was excited to a specific energy level and before the lifetime of this quantum state ends, another photon with energy (E_y) passes through it and a photon is emitted from the atom by stimulated emission with energy equal to that of helium-neon laser. So, the energies of the two photons (E_x) and (E_y) are ..... .



	E_x	E_y
(a)	$E_1 - E_0$	$E_1 - E_0$ ✗
(b)	$E_1 - E_0$	$E_2 - E_1$ ✓✓
(c)	$E_2 - E_0$	$E_2 - E_0$ ✗
(d)	$E_2 - E_0$ ✓✓	$E_2 - E_1$ ✓✓



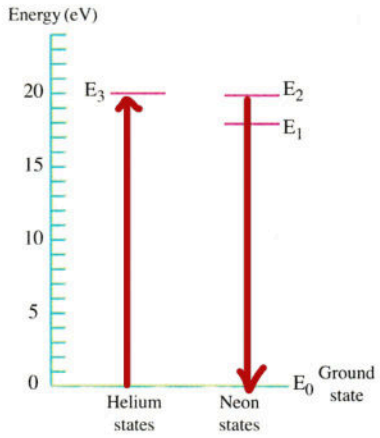
48 The opposite schematic diagram represents the levels of energy in helium and neon atoms, hence the largest energy difference between these levels is when moving between the two levels ..... .

(a) E_1 and E_3

(b) E_1 and E_2

(c) E_0 and E_1

(d) E_2 and E_0 or E_3 and E_0



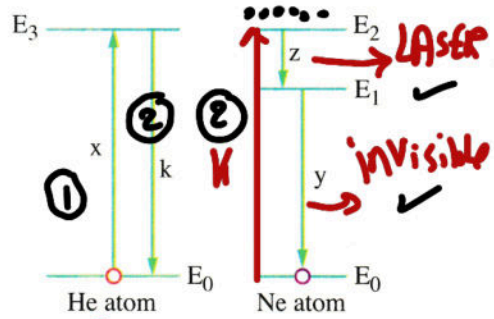
49 The opposite figure illustrates some transitions x, y, z and k in the helium-neon laser. The correct arrangement for these transitions according to the order of their occurrence is ..... .

(a) k ⟶ y ⟶ z ⟶ x

(b) x ⟶ y ⟶ z ⟶ k

(c) x ⟶ k ⟶ z ⟶ y

(d) z ⟶ y ⟶ x ⟶ k



Solve

50 The opposite figure represents the energy level diagram in a helium-neon laser, so:

(i) The energy required to excite helium atoms is ..... .

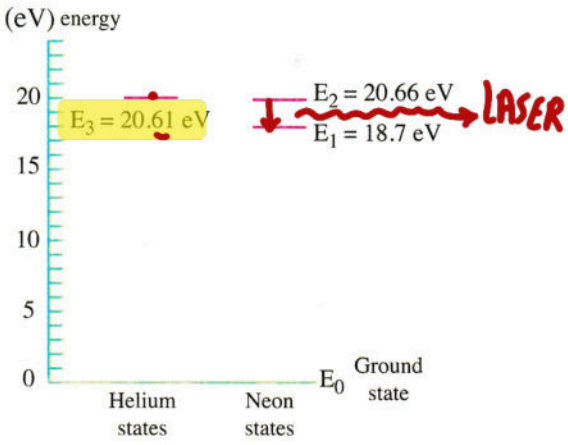
(a) 18.7 eV (b) 20.61 eV

(c) 20.66 eV (d) 21 eV

(ii) The energy of the emitted laser photons is ..... .

(a) 0.68 eV (b) 1.96 eV

(c) 3.12 eV (d) 4.05 eV



51 The opposite diagram represents the helium-neon laser device, so which part in the diagram represents the component in which the population inversion happens?

(a) (1). (b) (2).

(c) (3). (d) (1), (2).



Excitation

#Applications:

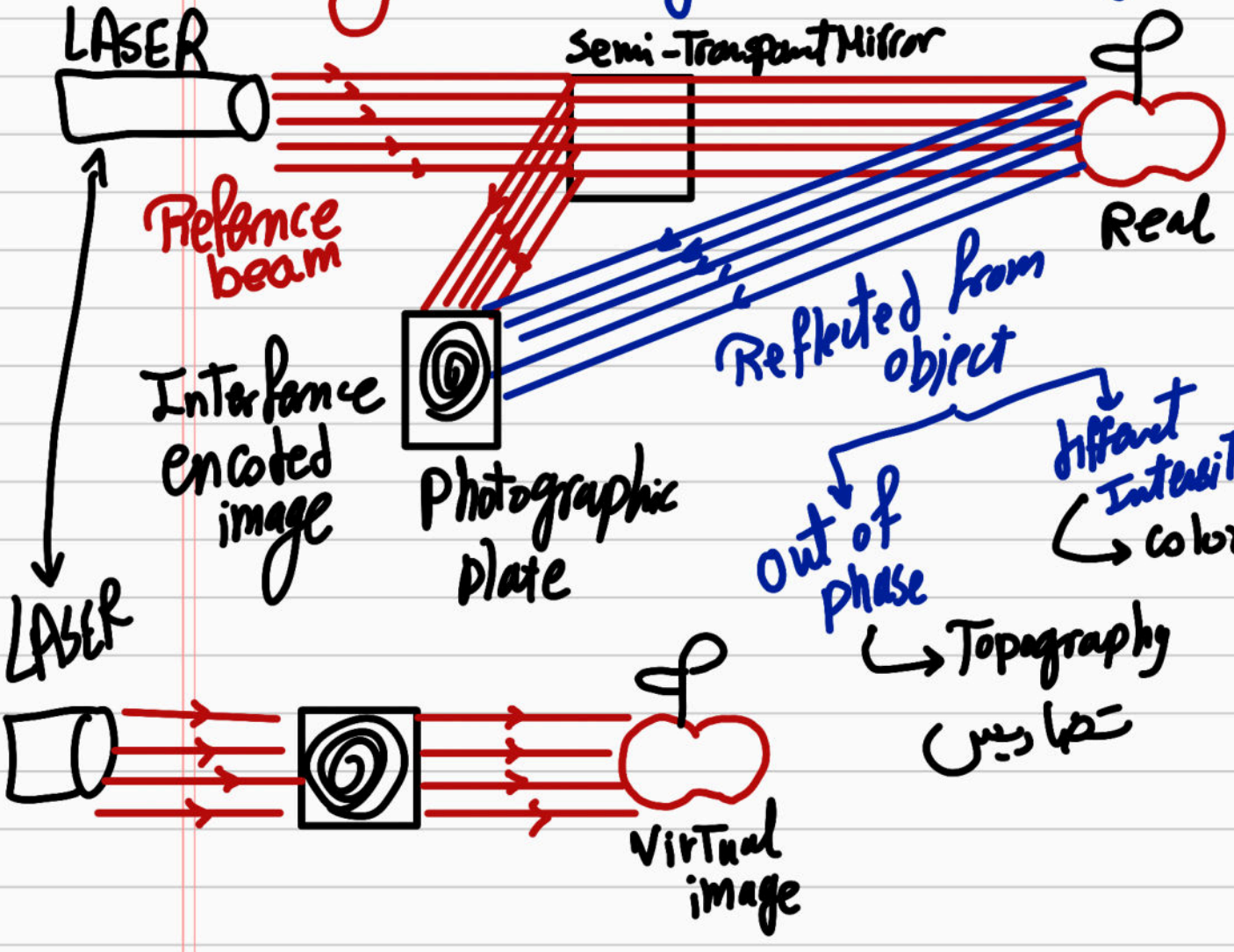
* Monochromatic [spectral purity]
  ↳ Light Expirments [Reflection / Refraction]

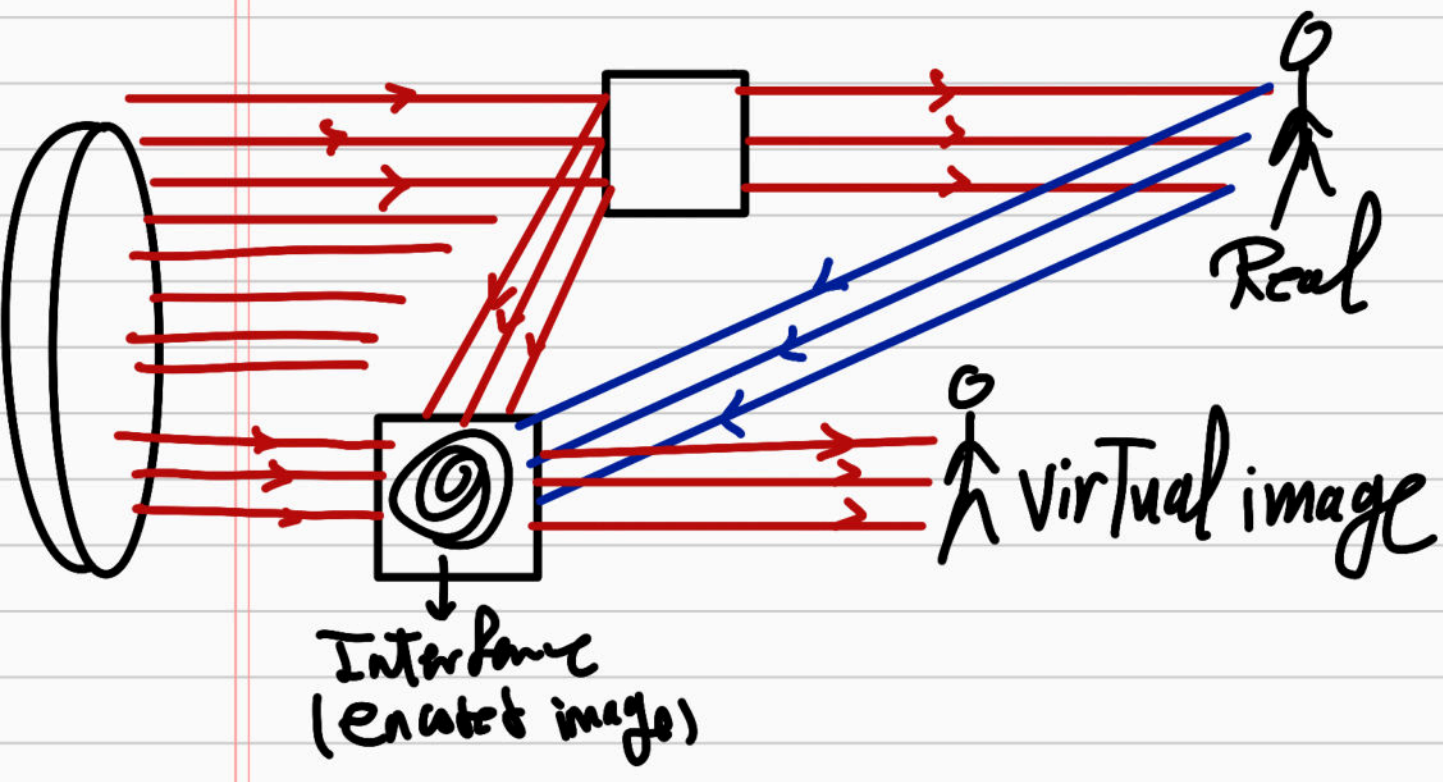
* Collimation [Parallel lines]
  ↳ Long distance applications
  → measurment
  → Miliatry
  Missiles (LADAR)



Thermal effat ← Medical field → Persion

* Cohenency → Hologram [3D-image]





* $\text{Phase diff} = \frac{2\pi}{\lambda} \times \text{Path diff.}$

* $\text{Path diff.} = \frac{\lambda}{2\pi} \times \text{Phase diff.}$

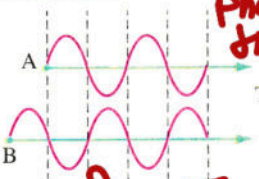
52 The characteristic of laser rays which allows using it in holograms is that ..... .

(a) their photons are coherent
(b) their rays are parallel
(c) they keep their intensity constant
(d) they have high intensity

53 A laser ray of wavelength λ is used in holography where the phase difference between two rays of the reflected rays from the body was $\frac{\pi}{2}$, then the path difference between them is .....

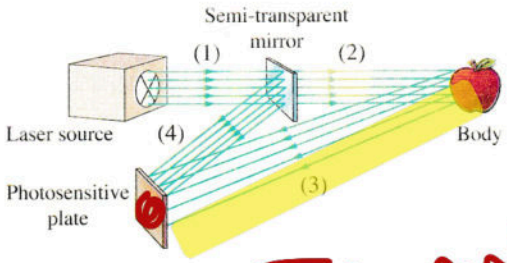
(a) $\frac{\lambda}{4}$ ✓
(b) $\frac{\lambda}{2}$
(c) 2λ
(d) 4λ

54 ✱ The opposite figure illustrates two laser waves reflected from two points A and B lying on a body during 3D hologram recording, if the path difference between them is 3164 Å, then the wavelength of the laser is .....



(a) 1582 Å
(b) 3164 Å
(c) 4746 Å
(d) 6328 Å ✓

55 The opposite figure illustrates how a photo of a body is recorded on a photosensitive plate, so:



(i) The set of rays that are different from each other in intensity is ..... .

(a) (1)
(b) (2)
(c) (3) ✓
(d) (4)

(ii) The set of rays that are different from each other in phase is .....

(a) (1)
(b) (2)
(c) (3) ✓
(d) (4)

(iii) The photo that is formed on the plate will be ..... .

(a) two-dimensional and similar to the body
(b) three-dimensional and similar to the body
(c) encoded into the form of interference fringes ✓
(d) a magnified photo similar to the body

56 The process of measuring long distances using laser rays depends on the property of ..... .

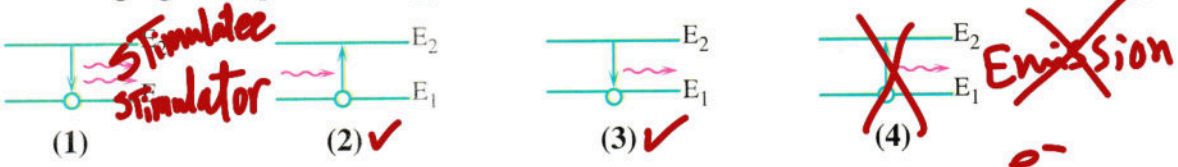
(a) the spectral purity of the laser
(b) the collimation of laser rays ✓
(c) the thermal effect of laser rays
(d) the high speed of laser rays

57 The laser is used in healing the retina detachment in the eye depending on ..... .

(a) the coherence of its photons ✓
(b) its thermal effect ✓
(c) its spectral purity ✓
(d) its high speed ✓

Second Essay questions

1. The following figures represent energy levels transitions of an atom:

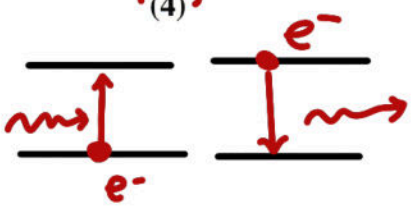


Show with explanation which of these figures represents:

(a) the absorption case?

(b) the stimulated emission case?

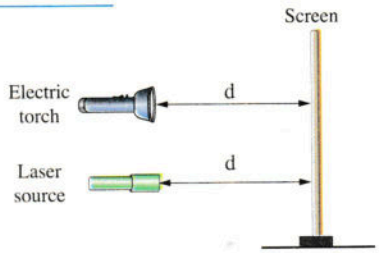
(c) the spontaneous emission case?



2. The stimulated emission process includes producing another coherent photon identical to the incident photon. **Is** getting these two photons considered a violation of the law of conservation of energy?

3. Laser beams don't obey the inverse square law. **Explain**.

4. The opposite figure shows a laser source and an electric torch having the same power and placed at the same distance from a screen, **compare between** the laser light intensity and the torch light intensity at the screen **explaining your answer**.



5. If the intensity of ordinary light at a distance of 1 m away from its source is 16 mW/m^2, **what** will be the intensity of that light at 2 m away from the source?

6. **In (helium-neon) laser, what is the function of each of:**

(a) Neon atoms?

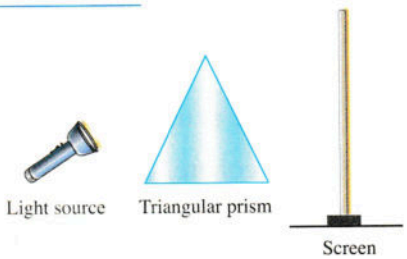
(b) The voltage source?

7. During the researches of producing laser, scientists have found that the laser efficiency increases very much when helium is added to neon instead of using neon only. **Explain** the role of helium in the laser production.

8. **In the opposite figure, describe with explanation the formed spectrum on the screen if the light source is:**

(a) Helium-neon laser.

(b) Tungsten lamp.



9 **How** does the laser beam get amplified in the helium-neon laser?

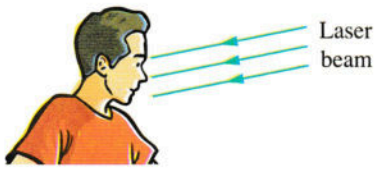
10 **Give reasons for:** An X-ray photon has higher energy than a helium-neon laser photon, although helium-neon laser rays can reach farther distances than X-rays.

11 **Mention the points of the similarities and the differences between** the photons of helium-neon laser and the photons of X-rays.

12 **Explain why** when a helium-neon laser beam falls on the surface of a metal, photoelectrons aren't emitted from the surface even if the beam has high intensity, but when a white light falls on the same surface, photoelectrons are emitted.

13 **What happens when** illuminating the holographic film with a laser beam that has the same wavelength as the reference beam and looking through it with bare eyes?

14 In the opposite figure, **how** will the reflected light rays from the body be different than the incident light rays on it?



15 **Mention one condition for** recording the topography of the object on a photographic plate.

16 **Why are laser rays used in each of the following:**

(a) Missile guidance in modern warfare applications?

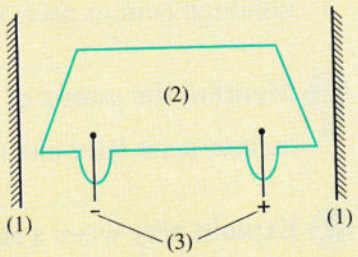
(b) Holography?

(c) Retina detachment treatment?

Questions of the Previous Years Exams On Chapter 7

1 The schematic diagram represents the apparatus of (He - Ne) laser production, which choice correctly represents the role of each of the components (1, 2 and 3)?

(Exp. June 21 - Exp.25)



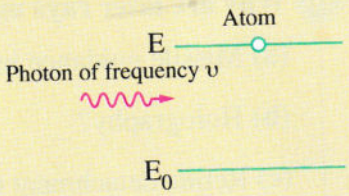
	Component 1	Component 2	Component 3
(a)	Photons' production	High voltage production	Reflecting photons
(b)	Reflecting photons	Containing the active medium	High voltage production
(c)	Pumping energy to excite the atoms	Exciting the neon atoms	Amplifying photons
(d)	Laser photons' production	The used energy source	Exciting the neon atoms

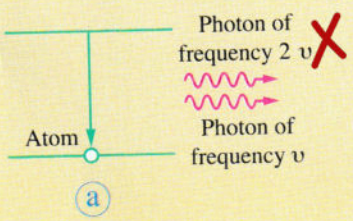
2 In ruby - chromium laser, strong xenon bulbs are used to excite the atoms of the active medium, so the ratio of $\frac{\text{the speed of produced laser ray in air}}{\text{the speed of xenon lamp light in air}}$ ..... *(Exp. June 21)*

(a) is more than one (b) equals one (c) is less than one (d) equals zero

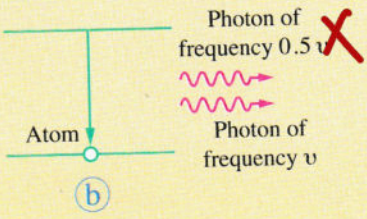
3 A photon of frequency υ falls on an excited atom as in the opposite figure, so the figure that represents the stimulated emission is figure number ..... .

(Exp. June 21)

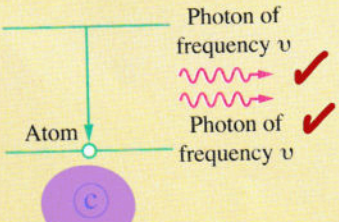




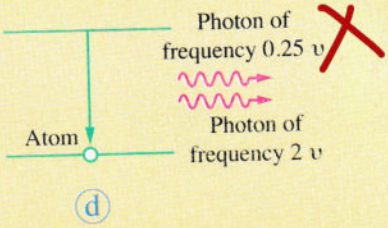
(a)



(b)

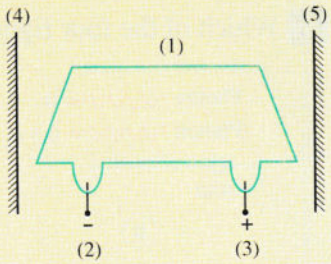


(c)



(d)

4 The opposite figure represents a schematic diagram of (He - Ne) laser, its components are (1, 2, 3, 4 and 5), what is the correct choice that plays an important role in amplifying the laser photons? *(1st session 21)*



Resonant Cavity (Mirrors)

(a) (1) and (2). (b) (4) and (5). (c) (1) and (4). (d) (3) and (5).

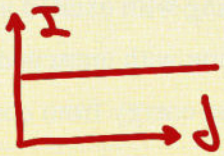
5 A laser beam has a diameter of 0.2 cm and an intensity of I at its source, then what happens to its diameter and intensity at 12 m apart from the source? *(1st session 21)*

(a) Both diameter and intensity remain constant. ✓

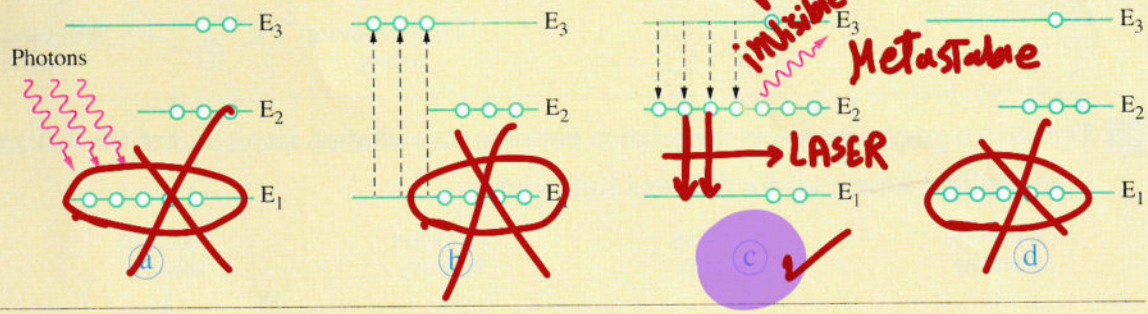
(b) The diameter increases while the intensity decreases.

(c) Both diameter and intensity increase.

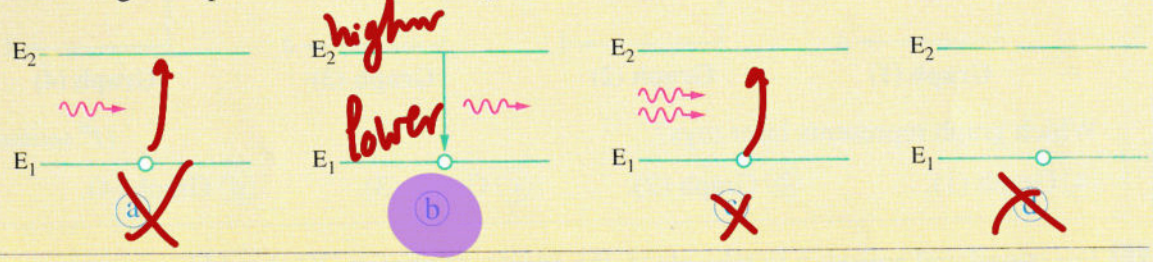
(d) Both diameter and intensity decrease.



6 Four of the laser generation stages are represented in the following diagrams, which of these figures represents population inversion? *(1st session 21)*



7 Which figure represents an emission process? *(1st session 21)*



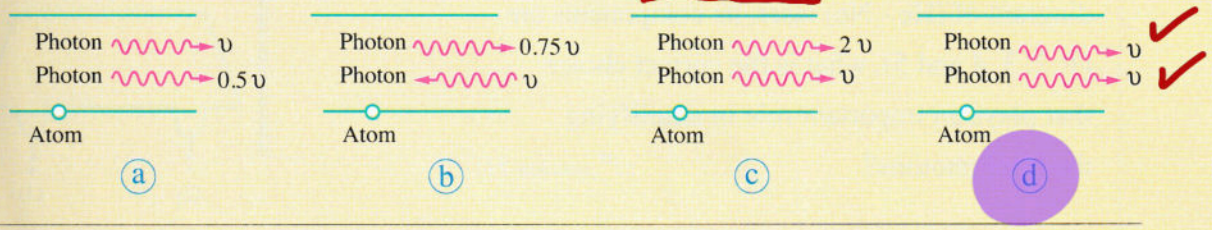
8 In a process of 3D imaging for an object using laser, if the path difference between two reflected laser waves from the object is $\frac{2}{3}\lambda$, the phase difference between these waves equals ..... . *(2nd session 21)*

(a) $\frac{3}{4}\pi$ (b) π (c) $\frac{4}{3}\pi$ ✓ (d) $\frac{3}{2}\pi$

$$\text{Phase} = \frac{2\pi}{\lambda} \times \text{Path} = \frac{2\pi}{\lambda} \times \frac{2}{3}\lambda = \frac{4\pi}{3}$$

LASER

9 Which of the four figures represents the concept of monochromaticity of laser? *(2nd session 21)*



(a) (b) (c) (d)

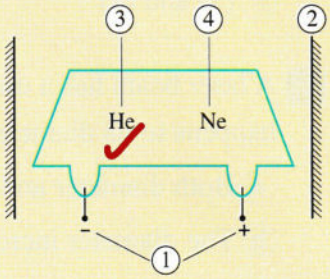
10 The schematic diagram represents the (He-Ne) laser, so neon (Ne) atoms can be excited by ..... . *(2nd session 21 , Exp.25)*

(a) colliding with component (2)

(b) colliding with excited atoms of component (3)

(c) colliding with unexcited atoms of component (3)

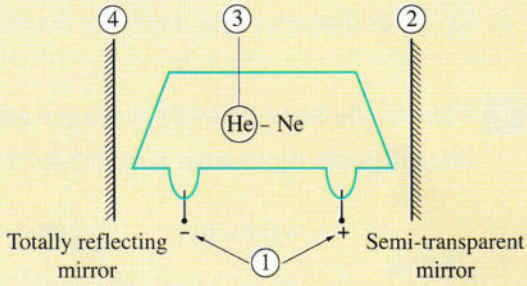
(d) gaining energy from component (1)



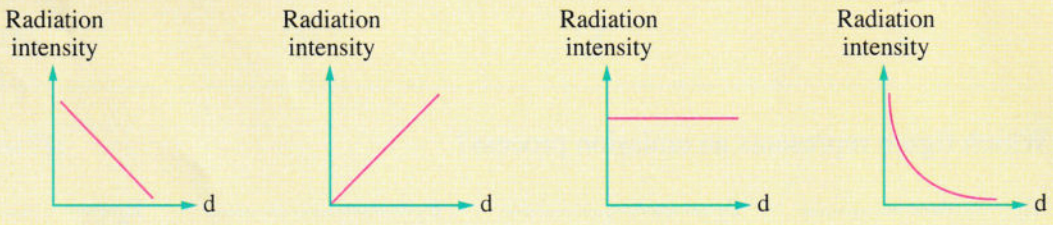
11 The diagram represents the (He - Ne) laser, which of the components (1, 2, 3 and 4) is responsible for neon atoms excitation? *(1st session 22)*

(a) 4 (b) 1

(c) 2 (d) 3



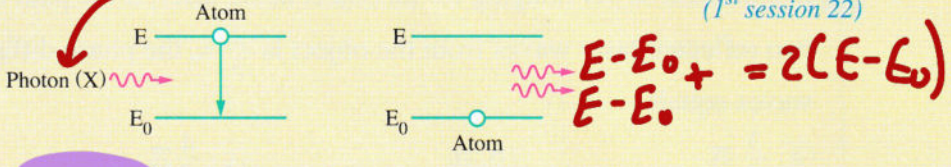
12 The shown graphs represent the relation between the radiation intensity and the distance from the light source (d) for different light sources.



Which graph represents laser rays? *(1st session 22)*

(a) Graph (1). (b) Graph (2). (c) Graph (3). (d) Graph (4).

13 For stimulated emission to take place, the energy of photon (X) should be = ... *(1st session 22)*



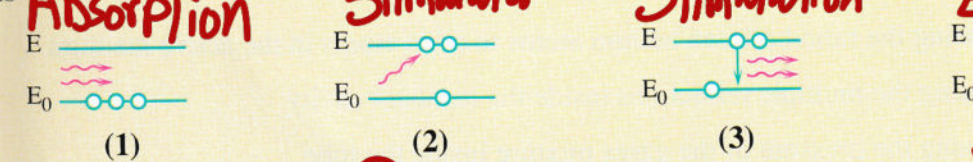
(a) $E + E_0$ (b) $E - E_0$ (c) $2(E - E_0)$ (d) $2(E + E_0)$

E-E0

(1) (4) (2) (3)

14 Using the following diagrams, the correct arrangement for the stages of laser production is ..... *(2nd session 22)*

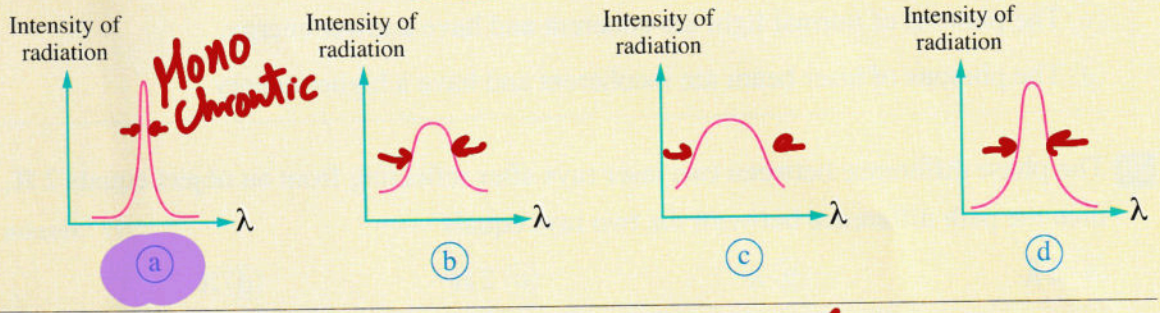
Absorption Stimulator Stimulation Excitation



(a) 1 → 2 → 4 → 3 ✓
(b) ✗ 2 → 1 → 4 → 3
(c) ✗ 4 → 1 → 2 → 3
(d) 1 → 4 → 2 → 3 ✓

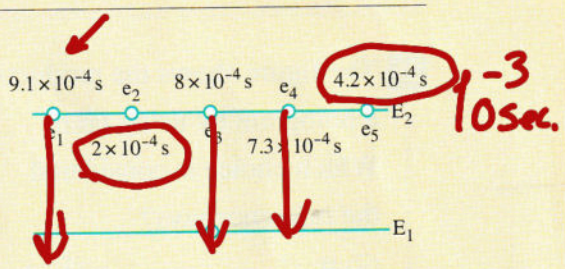
Population inversion

15 The shown graphs represent the relation between the radiation intensity and the wavelength (λ) for different light sources with the same scale. Which graph represents the source that can be used in 3D imaging? LASER *(2nd session 22)*



Mono Chromatic

16 The figure shows the population inversion of neon gas and the time interval that each of five excited atoms spent in the metastable energy level (E_2) until a certain instant. Assuming that 5×10^{-4} s later from that instant, a group of stimulating photons each of energy ($E_2 - E_1$) will reach the five excited atoms, which of these 5 atoms will be stimulated before elapsing their lifetime? (Assuming that the lifetime of the metastable energy level (E_2) = 10^{-3} s) *(2nd session 22)*



10^{-3} sec.

e_1 $9.1\times10^{-4} + 5\times10^{-4} = 1.4\times10^{-3}$ s after lifetime

(a) e_1, e_3 (b) e_2, e_4 (c) e_2, e_5 ✓ (d) e_1, e_2, e_5

17 On replacing one of the two mirrors in a laser device by a transparent glass piece and switching on the device, ... *(Exp. 23)*

Light ✓

(a) the laser beam comes out from the side of transparent glass piece
(b) the laser beam comes out from the side of the mirror
(c) no laser beam will be produced from the device ✓
(d) the laser beam comes out from the two sides of the device

Amplification

L. A. S. E. R.

18 The intensity of laser ray in (He-Ne) laser increases by ..... . *(Exp. 23)*

(a) decreasing the used potential difference inside the quartz tube

(b) decreasing the ratio between helium atoms to neon atoms in the active medium

(c) increasing the multiple reflections inside the resonant cavity

(d) increasing the pressure of the gases mixture inside the tube

19 Two sources, one of them is a normal light source emits monochromatic blue light and the second emits laser beam in red spectrum region. Which of the following sentences is correct? *(Exp. 23)*

(a) The photons of laser beam are coherent and have a higher energy.

(b) The photons of normal light are incoherent and have a higher energy.

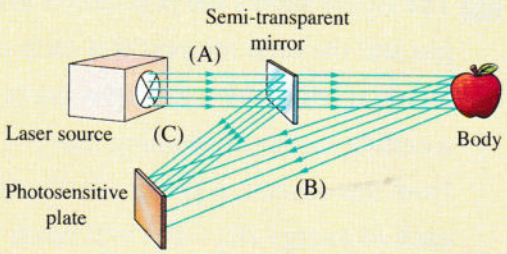
(c) The photons of normal light are coherent and have a lower energy.

(d) The photons of laser beam are incoherent and have a higher energy.

20 The phase difference between two laser rays after reflecting from an object equals 2π, then the path difference between the two rays equals ..... *(1st session 23)*

(a) 2λ (b) λ (c) 2π (d) π

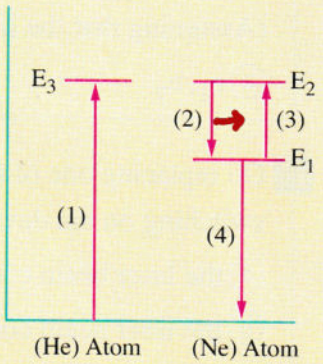
21 The figure shows the production of a hologram by using laser. Which choice is considered the reference rays? *(1st session 23)*



(a) B, C (b) A, B

(c) Only C (d) Only B

22 The figure illustrates laser photon production from the mixture of two gases (He) and (Ne), knowing that both levels E_2 and E_3 are metastable levels, which transmission causes a laser photon emission? *(1st session 23)*



(a) Transition (4).

(b) Transition (3).

(c) Transition (2).

(d) Transition (1).

23 One of the millitary applications of laser is for destroying missiles this is due to ..... .

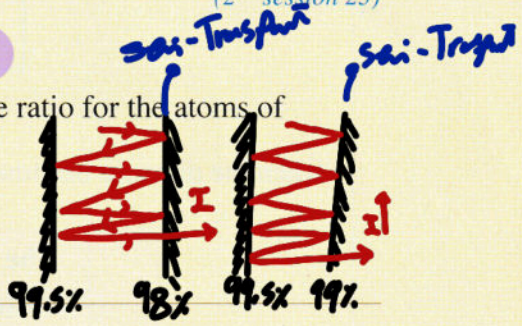
(2nd session 23)

(a) the wave nature of laser beam
(b) the coherency of laser beam photons
(c) the energy of laser beam
(d) the monochromaticity of laser beam

24 The emergence of laser beam from the semi-transparent mirror in He-Ne laser starts depending on ..... .

(2nd session 23)

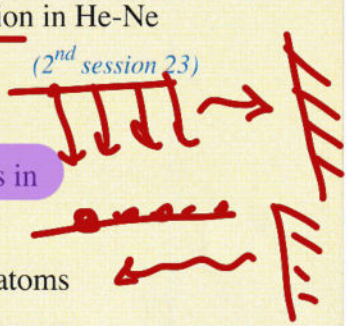
(a) the intensity of the radiation inside the resonant cavity
(b) achieving the state of population inversion with a definite ratio for the atoms of the active medium
(c) the electric potential difference inside the laser tube
(d) the lifetime of the metastable level



25 The amplification of the number of photons produced by stimulated emission in He-Ne laser is due to the collisions of ..... .

(2nd session 23)

(a) the excited neon atoms in the metastable level with each other
(b) the reflected photons in the resonant cavity with the excited neon atoms in the metastable level
(c) the excited neon atoms in the metastable level with the excited helium atoms
(d) the excited helium atoms with the unexcited neon atoms



26 In holography, which of the following light beams has a phase difference between its photons?

(1st session 24)

(a) The beam emitted from the source and falls on the mirror.
(b) The beam emitted from the source and falls on the object.
(c) The reflected beam from the mirror to the photographic plate.
(d) The reflected beam from the object to the photographic plate.

27 If the phase difference between the rays used in holography is equal to $\frac{\pi}{4}$, which of the following choices represents the path difference between the rays?

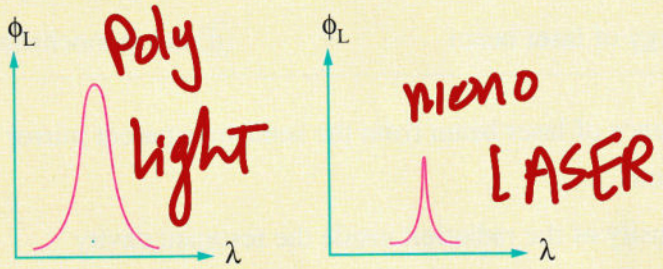
(1st session 24)

(a) $\frac{\lambda}{2}$
(b) $\frac{\lambda}{4}$
(c) $\frac{\lambda}{8}$
(d) $\frac{\lambda}{16}$

$$\text{Path} = \frac{\lambda}{2\pi} \times \frac{\pi}{2} = \frac{\lambda}{8}$$

28 Two light sources spectra are shown in the figures, if the intensity of light emitted from the source (1) after a distance (d) is 2I and the intensity of light emitted from the source (2) after a distance (d) is I,

$\frac{I_1}{I_2} = \left(\frac{d_2}{d_1}\right)^2$



(1) **(2)**

when the distance becomes (2d), so the intensity becomes ..... . *(1st session 24)*

	The intensity of light emitted from source (1) at 2d	The intensity of light emitted from source (2) at 2d
(a)	$\frac{I}{4}$	2I
(b)	$\frac{I}{2}$ ✓	I ✓
(c)	2I	$\frac{I}{4}$
(d)	$\frac{I}{4}$	I

29 In He-Ne laser by replacing the semi-transparent mirror by a transparent glass plate, which of the following choices is correct? *(2nd session 24)*

(a) The intensity of laser beam increases to max. value.

(b) No stimulated emission takes place at all.

(c) No laser beam is produced at all. ✓

(d) No population inversion takes place at all.

30 Laser is used in 3D imaging (Holography) as its rays are characterized by ..... . *(2nd session 24)*

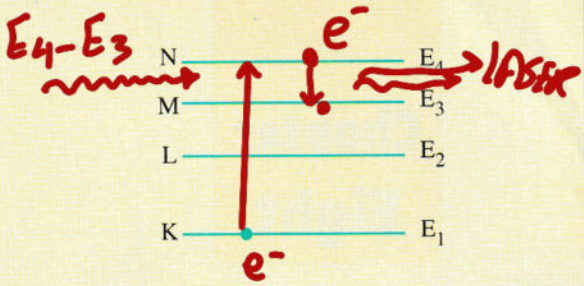
(a) high intensity radiation

(b) the coherency of photons ✓

(c) the monochromaticity of photons

(d) the effect on photographic plates

31 The figure shows an electron in the first energy level for a certain atom. If a photon of energy $E = E_4 - E_1$ fell on the electron, and before ending the lifetime of the electron in the level another photon of energy $E = E_4 - E_3$ fell on the excited electron, which choice is correct? *(2nd session 24)*



(a) The electron returns from N to K level by stimulated emission.

(b) The electron returns from N to M level by spontaneous emission.

(c) The electron returns from N to M level by stimulated emission.

(d) The electron returns from N to K level by spontaneous emission.

32 The spectral purity of laser beams means that their photons ..... *(Exp. 25)*

(a) have one direction

(b) have approximately one wavelength

(c) are in phase

(d) do not follow the inverse square law

33 The form of energy used to excite the atoms of the active medium in liquid dye lasers is ..... energy.

(a) optical (b) electrical (c) thermal (d) chemical *(Exp. 25)*

34 Helium and neon are chosen as the active medium for producing a (He-Ne) laser because ..... . *(Exp. 25)*

(a) they have the same number of energy levels

(b) the values of the stable excited energy levels in both are close

(c) the values of the metastable excited energy levels in both are close

(d) all of the above

35 In the (He-Ne) laser, when the semi-transparent mirror (reflectivity 95 %) is replaced with another (reflectivity 98 %) the produced laser intensity ..... and the energy of each photon from the produced laser ..... respectively. *(Exp. 25)*

(a) changes, remains constant

(b) remains constant, remains constant

(c) remains constant, changes

(d) changes, changes